

Hands-on Lab: Generative AI Applications for IT Professionals



Estimated time needed: 30 minutes

Welcome to the hands-on lab Generative AI Applications for IT Professionals!

Introduction

Generative AI (Gen AI) holds exciting potential for IT professionals across various roles, including software developers, IT security professionals, and web developers. IT professionals can leverage generative AI in various ways to enhance their work and deliver more efficient and innovative solutions.

In this lab, you will explore the capabilities of generative AI to demonstrate a few use cases for software developers, IT security professionals, and web developers.

The lab is designed in the Generative AI Classroom environment, and you will work with the gpt-4o-mini model.

Learning objectives

After completing this lab, you will be able to:

- Demonstrate examples of the use of generative AI by software developers.
- Demonstrate examples of the use of generative AI by IT security professionals.
- Demonstrate examples of the use of generative AI by web developers.

Set up the AI classroom

The generative AI classroom is a tool where you can write and compare your prompts to generate desired text with real-time chat responses. Moreover, you can choose from multiple generative AI models and learn about their strengths and weaknesses.

You will have both the generative AI classroom environment and instructions on one page in a browser. The instructions will be on the left half of the screen, and the generative AI classroom will be on the right half of the screen. You can interact with the language model using the message and chat fields.



Hands-on Lab: Generative AI Applications for IT Professionals



Skills
Network

Estimated time needed: 30 minutes

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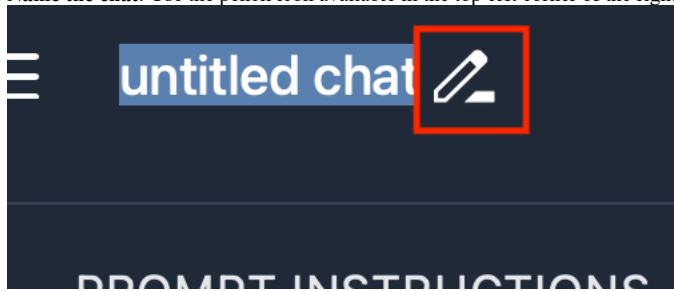
Set up the AI classroom

The generative AI classroom is a tool where you can write and compare your prompts to generate desired text with real-time feedback.

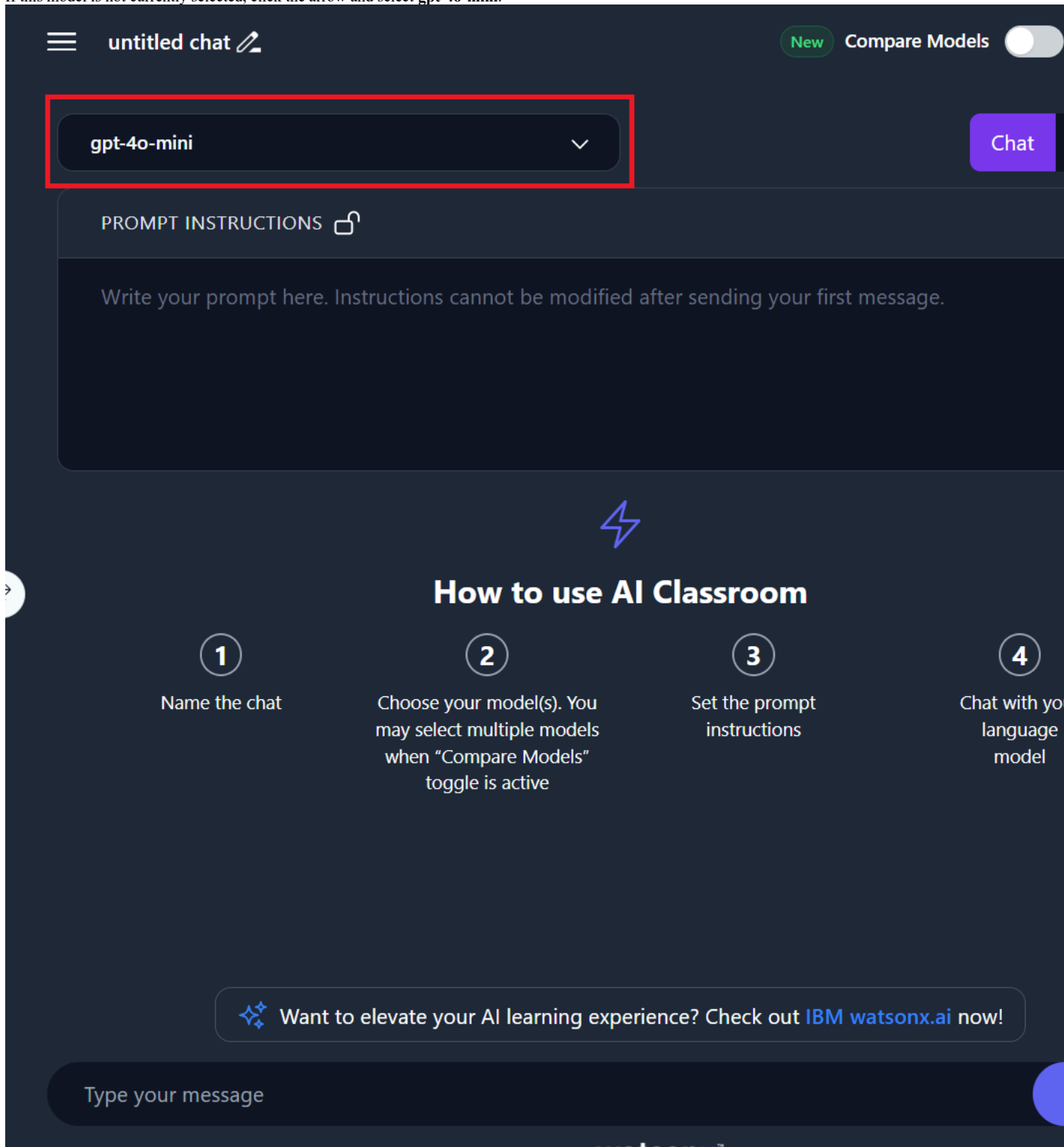
As a first step, you must set up your AI classroom for a better learning experience.

Before beginning the exercise, you must:

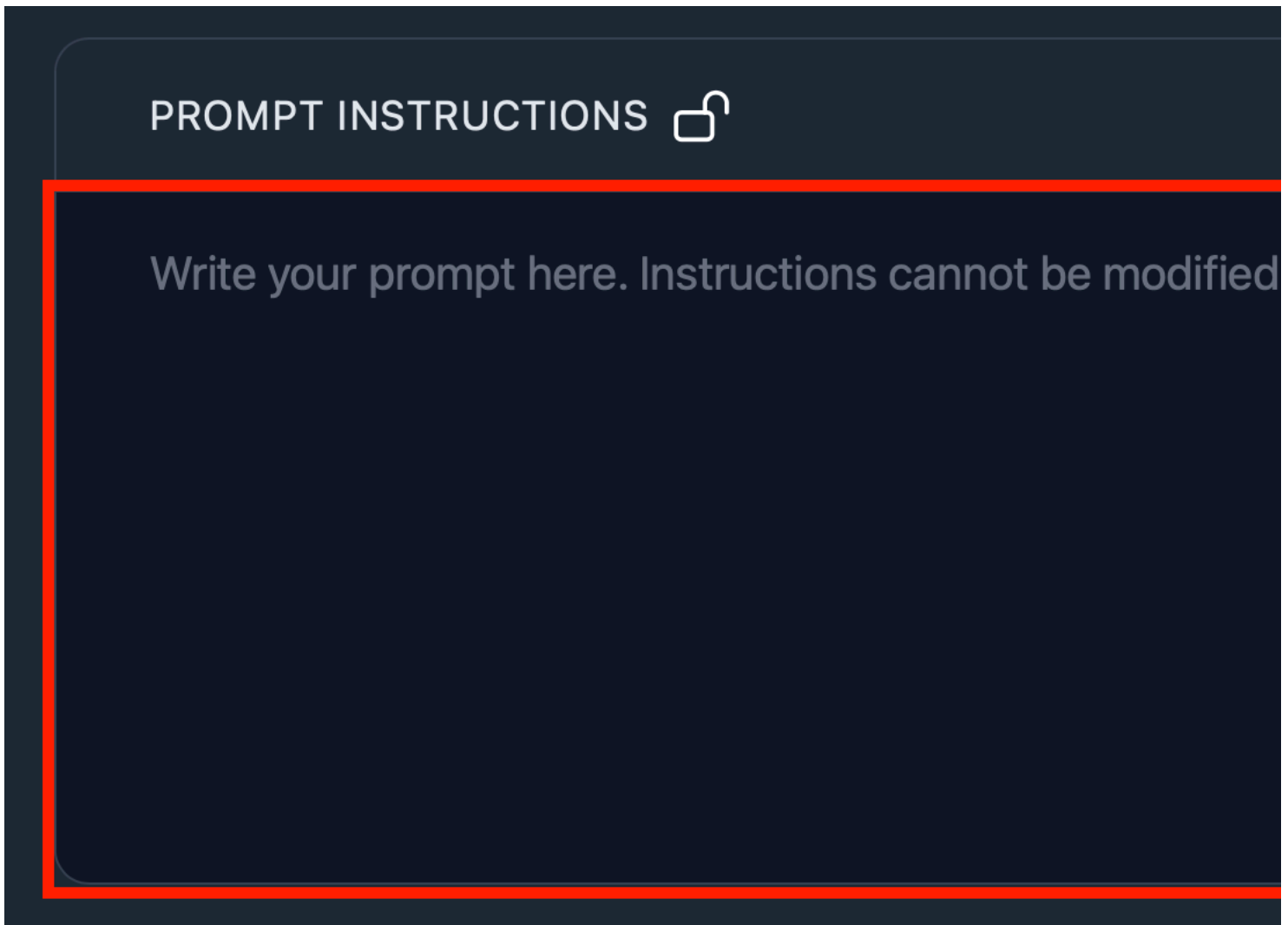
1. **Name the chat:** Use the pencil icon available in the top-left corner of the right pane to name the chat.



2. **Choose the model:** Use the dropdown option from the top-right corner of the right pane to select the model in your chat. The default selected model is gpt-4o-mini. If this model is not currently selected, click the arrow and select **gpt-4o-mini**.



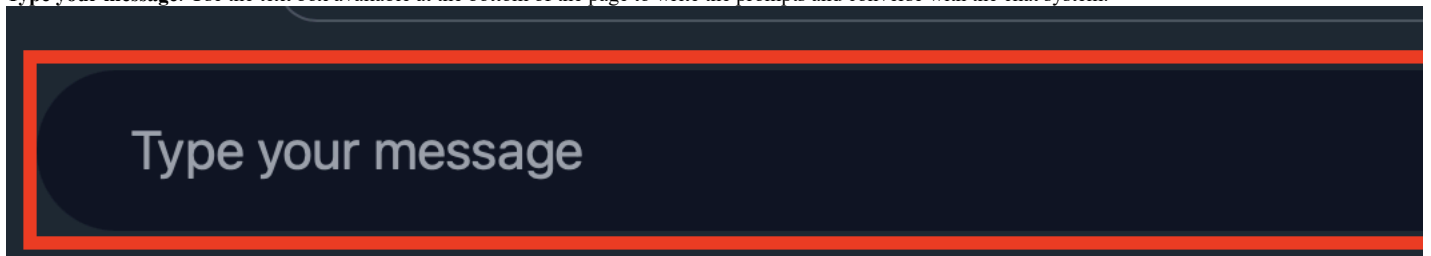
3. **Give prompt instructions:** Use the **Prompt Instructions** field on the upper right pane of the chat system to provide instructions or any specific details about the context of the required output. These instructions will be locked when you start the chat and cannot be modified later.



PROMPT INSTRUCTIONS 

Write your prompt here. Instructions cannot be modified

4. **Type your message:** Use the text box available at the bottom of the page to write the prompts and converse with the chat system.



Type your message

Generating and optimizing code

Software developers frequently experience time constraints and resource limitations that affect their overall efficiency and productivity. Manual responsibilities such as code documentation, knowledge management, and troubleshooting absorb precious time that could be more effectively utilized for innovation and proactive problem-solving. The challenge lies in effectively connecting technical expertise with user comprehension, which can be demanding and result in communication breakdowns and frustration. Conveying intricate technical terminology in a user-friendly manner poses a difficulty for software professionals, influencing support efficiency and user satisfaction.

Many generative AI models, such as GPT, can be used to generate code snippets or entire programs in different programming languages based on natural language descriptions or requirements. For a given problem or requirement, generative AI models can produce different code implementations, offering exploration of alternative solutions and potential optimization. Generative AI models can also be useful for code optimization by analyzing existing code and suggesting potential optimizations for performance, readability, or maintainability.

Step 1: Generating the code

1. Let's consider prompting the model to generate a Python function that calculates the Fibonacci sequence. Write your prompt or statement in the **Type your message** field at the bottom of the screen.

Create a Python function that calculates the Fibonacci sequence.

You can copy the example using the copy icon on the right side of the box and paste it into the Type your message field.

**Generating and optimizing code** 

Chat

Freeform

gpt-4o-mini 



PROMPT INSTRUCTIONS 

Write your prompt here. Instructions cannot be modified after sending your first message.

How to use AI Classroom

1

Name the chat

2

Choose your model

3

Set the prompt instructions

4

Chat with your language

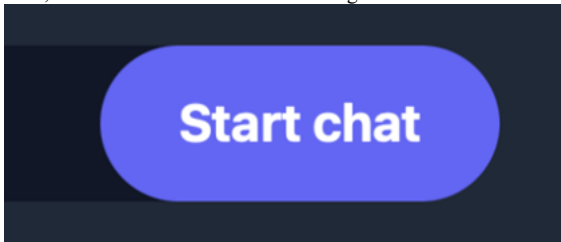
 Want to elevate your AI learning experience? Check out [IBM watsonx.ai](https://ibm.com/watsonx.ai) now!

Create a Python function that calculates the Fibonacci sequence.

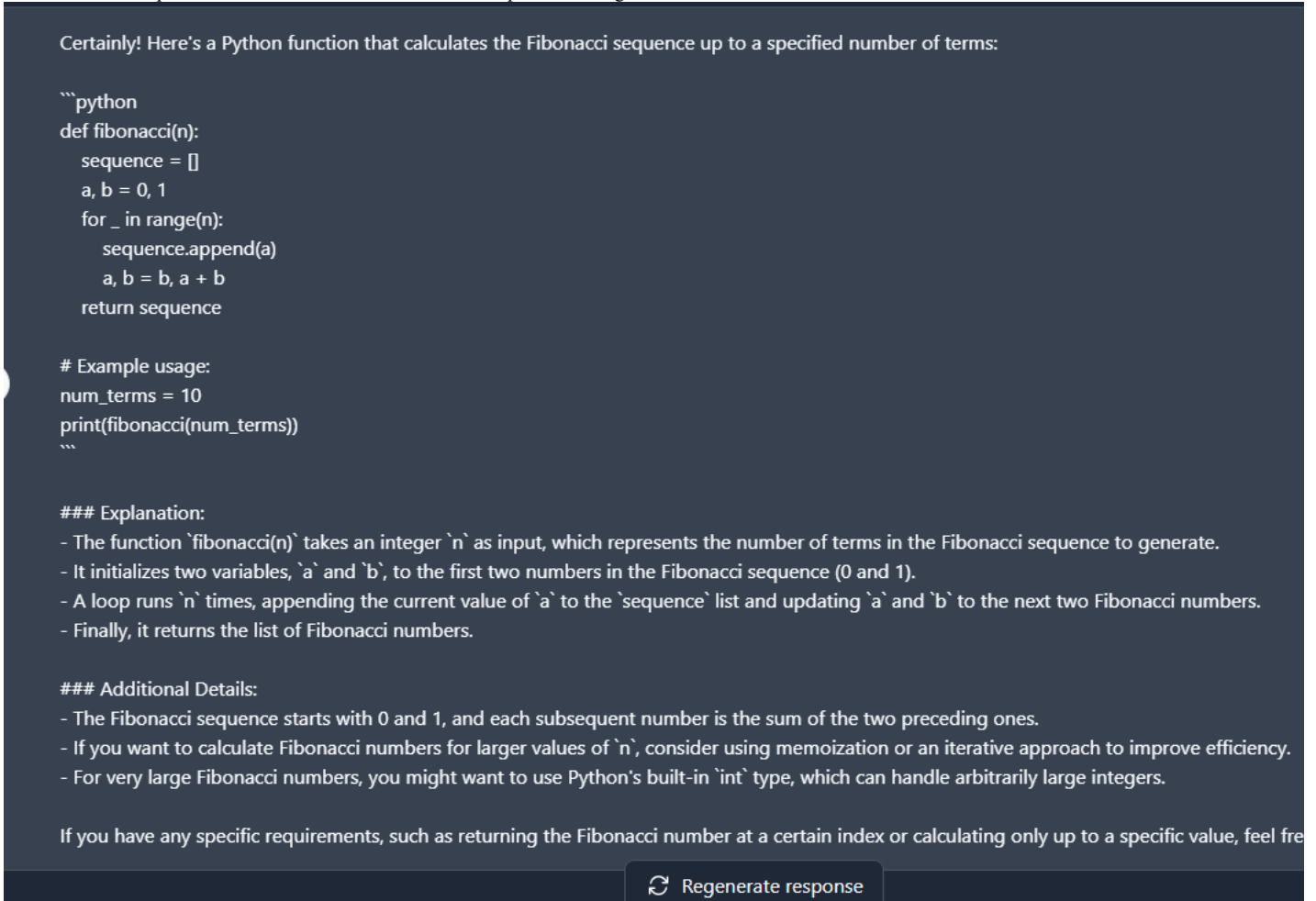
Start

Note: Ignore the prompt instructions field for easy understanding and a quick start. You can include additional information or instructions about the context of the required output along with your prompt in the type your message field.

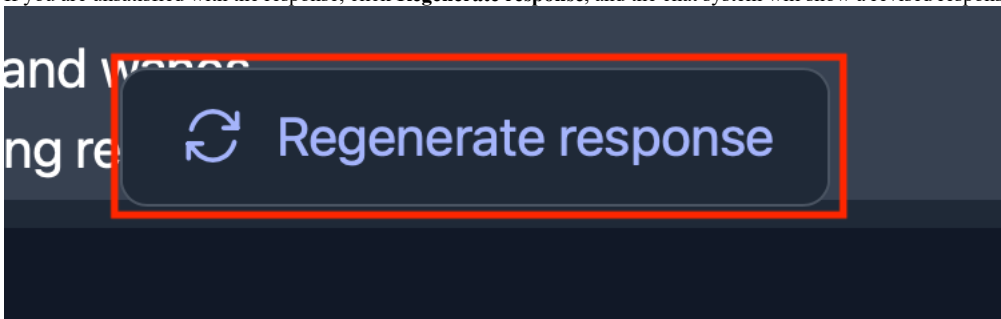
2. Now, click **Start chat** to send the message to the chatbot.



3. You will see the response with the desired content in the middle pane on the right side of the screen.



4. If you are unsatisfied with the response, click **Regenerate response**, and the chat system will show a revised response for your original prompt.



5. You can copy and paste the generated output and tweak it further as required.

Step 2: Optimizing the code

You can further prompt the model to refactor the generated code to improve readability, maintainability, or performance.

1. Continue with the same Step 1 example in the same chat window.
2. Firstly, let's enter the following prompt to add inline comments to improve readability and include an explanation of the code statements. Enter the additional prompt in the **Start chat** field.

Add inline comments to the code statements to provide explanations and improve readability.

You can copy the prompt using the copy icon on the right side of the box.

 **Generating and optimizing code** 

Chat

Freeform

gpt-4o-mini





PROMPT INSTRUCTIONS 

Instructions cannot be modified after sending your first message.

sequence to generate.

- It initializes two variables, `a` and `b`, to the first two numbers in the Fibonacci sequence (0 and 1).
- A loop runs `n` times, appending the current value of `a` to the `sequence` list and updating `a` and `b` to the next two Fibonacci numbers.
- Finally, it returns the list of Fibonacci numbers.

Additional Details:

- The Fibonacci sequence starts with 0 and 1, and each subsequent number is the sum of the two preceding ones.
- If you want to calculate Fibonacci numbers for larger values of `n`, consider using memoization or an iterative approach to improve efficiency.
- For very large Fibonacci numbers, you might want to use Python's built-in `int` type, which can handle arbitrarily large integers.

If you have any specific requirements, such as returning the Fibonacci number at a certain index or calculating only up to a specific value, feel free to ask!

 Regenerate response

Add inline comments to the code statements to provide explanations and improve readability.

3. Click the **Send** button. The chat system will generate the code with inline comments explaining the code statements.

```
Certainly! Here's the modified version of the fibonacci_at_index function with inline comments for clarity:
```

```
```python
def fibonacci_at_index(n):
 # Handle the case for negative indices
 if n < 0:
 return "Invalid input" # Return an error message for invalid input
 # Start with the first two Fibonacci numbers
 a, b = 0, 1
 # Loop until we reach the desired index
 for _ in range(n):
 # Update a and b to the next two Fibonacci numbers
 a, b = b, a + b
 # Return the Fibonacci number at the specified index
 return a

Example usage:
index = 10 # Specify the index to get the Fibonacci number
print(fibonacci_at_index(index)) # Print the Fibonacci number at the specified index
```
```

Explanation of Inline Comments:

- Each comment describes the purpose of the code that follows, making it easier to follow the logic.
- The comment for the error handling clarifies what happens if a negative index is provided.
- Comments within the loop explain how the Fibonacci numbers are updated.

Additional Details:

- This function efficiently computes the Fibonacci number at a specific index using an iterative approach, which is more efficient than a recursive method for larger indices.
- If you're interested in optimizing further for very large indices, you might consider using matrix

4. You can further prompt the model to optimize this prompt for performance. Let's enter the following prompt.

Optimize this code for performance.

You can copy the prompt using the copy icon on the right side of the box.

5. The chat system will respond to your additional message(s) based on already generated responses. The model has produced an optimized version of the code, with the related explanation.

To optimize the Fibonacci sequence calculation for performance, you can use a technique called memoization, which stores previously computed values to avoid redundant calculations. This is particularly useful for recursive implementations. However, for an iterative approach, we can reduce the space complexity by only storing the last two Fibonacci numbers instead of the entire sequence. Here's the optimized code:

```
```python
def fibonacci(n):
 """Generate the nth Fibonacci number using an optimized approach."""
 if n <= 0: # Handle cases where n is less than or equal to 0
 return []
 elif n == 1: # If n is 1, return the sequence with only the first number
 return [0]
 elif n == 2: # If n is 2, return the first two Fibonacci numbers
 return [0, 1]

 sequence = [0, 1] # Initialize the sequence with the first two Fibonacci numbers
 a, b = 0, 1 # Start with the first two Fibonacci numbers

 for _ in range(2, n): # Loop from the 3rd term to the nth term
 a, b = b, a + b # Update a and b to the next two Fibonacci numbers
 sequence.append(b) # Append the new Fibonacci number to the sequence

 return sequence # Return the complete Fibonacci sequence

Example usage:
num_terms = 10 # Specify the number of terms to generate
print(fibonacci(num_terms)) # Print the Fibonacci sequence of the specified number of terms
```
```

Explanation of Optimizations:

- **Space Complexity**: The function now

 Regenerate response

quires only two Fibonacci numbers. The initial

Identifying data security measures

The IT security team can leverage generative AI for diverse use cases, including anomaly detection, analysis of potential security vulnerabilities, and designing security policies.

In this exercise, we will demonstrate the use of generative AI for identifying data security measures for an organization. For this example, let's consider a fictitious mid-size enterprise called TechData, which has a growing, diverse IT infrastructure, including on-premises servers, cloud services, and a distributed workforce. TechData deals with sensitive customer data, financial records, and proprietary business information. The IT security team is tasked with regularly updating and enhancing security policies to meet compliance requirements.

Let's see how the IT security team can leverage guidance from ChatGPT to identify data security and protection measures.

Identify data security and protection measures

1. Firstly, create a new chat by clicking the New Chat () icon. You can name the chat using the pencil icon available in the top-left corner.

2. Considering the context, design the relevant prompt. You should ensure that your prompt is descriptive to guide the model to produce the desired output but does not share any sensitive or confidential data. Let's enter the following prompt in the **Type your message** field.

TechData is a mid-size enterprise with a growing, diverse IT infrastructure, including on-premises servers, cloud services, and a distributed workforce. They deal with sensitive customer data, financial records, and proprietary business information. The IT security team regularly updates and enhances security policies to meet compliance requirements. Suggest and prioritize security and data protection measures, including encryption standards and access controls.

☰ Data security and protection measeures ✎

Chat

Freeform

gpt-4o-mini ▼

PROMPT INSTRUCTIONS 📋

Write your prompt here. Instructions cannot be modified after sending your first message.

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TechData is a mid-size enterprise with a growing, diverse IT infrastructure, including on-premises servers, cloud services, and a distributed workforce. The organization must adhere to industry-specific compliance standards and regulations to ensure the security of sensitive data. They deal with sensitive customer data, financial records, and proprietary business information. The IT security team regularly updates and enhances security policies to meet compliance requirements. Suggest and prioritize security and data protection measures, including encryption standards and access controls.

Start

You can copy the prompt using the copy icon on the right side of the box.

3. Click the **Start chat** button. The model will produce suggestions regarding the security and data protection measures for TechData.

For TechData, as a mid-size enterprise with a diverse IT infrastructure and compliance requirements, it is crucial to prioritize security and data protection measures. Here are some suggestions:

1. **Implement strong encryption standards:** Ensure that sensitive data, such as customer information and financial records, is encrypted both at rest and in transit. Use industry-standard encryption algorithms like AES-256 for maximum security.
2. **Enforce access controls:** Implement a robust access control mechanism to restrict unauthorized access to sensitive data. This can include role-based access control (RBAC), multi-factor authentication (MFA), and regular access reviews.
3. **Regularly update and patch systems:** Keep all on-premises servers, cloud services, and software up to date with the latest security patches. This helps protect against known vulnerabilities and reduces the risk of exploitation.
4. **Implement data loss prevention (DLP) solutions:** Deploy DLP solutions to monitor and prevent the unauthorized transmission of sensitive data. This can include monitoring network traffic, email attachments, and USB devices for potential data leaks.
5. **Conduct regular security assessments:** Perform periodic security assessments to identify vulnerabilities and gaps in the IT infrastructure. This can include penetration testing, vulnerability scanning, and security audits.
6. **Train employees on security best practices:** Educate the distributed workforce on security awareness and best practices. This can include training on phishing attacks, password hygiene, and social engineering prevention.
7. **Implement a robust incident response plan:** Develop and regularly test an incident response plan to effectively respond to security incidents. This includes procedures for containment, investigation, and recovery.

By prioritizing these security and data protection measures, TechData can enhance its overall security posture and ensure compliance with industry-specific standards and regulations.

4. If you are unsatisfied with the response, click **Regenerate response**, and the chat system will show a revised response for your original prompt.

Leveraging generative AI for web development

A (fictitious) small web development company, WebTech Solutions, is working on creating a new website for a (fictitious) local restaurant, "Taste & Delicacy." In such a case, the developers can leverage generative AI for diverse use cases, such as:

- Generating a preliminary sitemap and content structure
- Generating website design mockups
- Translate the approved design mockups into HTML, CSS, and JavaScript code
- Generate variations of text for different sections (for example, the header, home page, menu descriptions, and blog posts) based on the restaurant's brand voice and target audience

Let's experiment with some simple use cases.

Identify the layout, color scheme, and content for the website

1. Firstly, create a new chat by clicking the New Chat ()icon. You can name the chat using the pencil icon available in the top-left corner.
2. Let's prompt the model to suggest some layouts and color schemes for the website. Enter the following prompt.

A small web development company, WebTech Solutions, is working on a new website for a local restaurant, Taste & Delicacy, which wants to blend tradition and modernity. Suggest some layouts and color schemes for the website.



Generative AI for web development

Chat

Freeform

gpt-4o-mini



PROMPT INSTRUCTIONS

Write your prompt here. Instructions cannot be modified after sending your first message.

How to use AI Classroom



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A small web development company, WebTech Solutions, is working on a new website for a local restaurant, Taste & Delicacy, which wants to blend traditional recipes with modern techniques, emphasizing both the tradition and innovation behind the dishes.
Suggest some layouts and color schemes for the website.

Start

powered by **watsonx**

3. Next, click **Start chat** to send the message to the chatbot. The output produced at our end is shown in the image.

For the website of Taste & Delicacy, blending traditional recipes with modern techniques, here are some layout and color scheme suggestions:

Layout Ideas:

1. **Hero Image Layout:**

- **Top Section:** A full-width hero image showcasing a signature dish, overlaid with a tagline that captures the essence of the restaurant.
- **Scrolling Sections:** Below the hero, use a parallax scrolling effect to highlight different sections such as "Our Story," "Menu," "Chef's Specials," and "Reservations."

2. **Grid Layout:**

- **Menu Grid:** Present the menu items in a grid format with images, descriptions, and prices. Hover effects could show additional details or cooking techniques used.
- **Recipe Spotlight:** Feature a section that highlights a traditional recipe alongside its modern interpretation, using a two-column layout.

3. **Card-Based Layout:**

- **Blog/News Section:** Use cards for blog posts or news updates about special events, new dishes, or culinary techniques, allowing easy navigation and readability.
- **Testimonials:** Create a card layout for customer reviews, featuring images of diners enjoying their meals.

4. **Split Screen Layout:**

- **Tradition vs. Innovation:** Use a split-screen design that juxtaposes traditional dishes on one side and modern interpretations on the other, with a clear call-to-action to explore both.

Color Schemes:

1. **Warm Earth Tones:**

- **Primary Colors:** Deep burgundy, olive green, and warm beige.
- **Accent Colors:** Gold or copper for buttons and highlights to evoke a sense of tradition and warmth.

2. **Modern Minimalist:**

- **Primary Colors:** Soft white and charcoal gray for a clean look.
- **Accent Colors:** Bright colors like teal or mustard yellow to highlight modern dishes and features, creating a fresh and inviting atmosphere.

3. **Rustic and Cozy:**

- **Primary Colors:** Rich browns and muted greens to reflect the warmth of traditional cooking.

4. Let's further prompt the model to suggest the sections and the related content for this website.

Suggest different sections and related content that can be included on this website.

5. Click the **Send** button to generate the output. The output generated at our end is shown in the image.

For the Taste & Delicacy website, here are some suggested sections along with related content that can enhance the user experience and effectively showcase the restaurant's unique blend of traditional recipes and modern techniques:

Suggested Sections and Related Content:

1. **Home:**

- **Welcome Message:** A brief introduction to the restaurant's philosophy, emphasizing the blend of tradition and innovation.
- **Featured Dish:** Highlight a special dish of the week with a mouth-watering image and description.
- **Call-to-Action Buttons:** Links to the menu, reservations, and special events.

2. **About Us:**

- **Our Story:** A narrative about the restaurant's origins, inspiration, and commitment to culinary excellence.

3. **Menu:**

- **Full Menu:** Comprehensive listing of all dishes, categorized by appetizers, main courses, desserts, and drinks.
- **Seasonal Specials:** Highlight seasonal or limited-time offerings that showcase fresh ingredients.
- **Dish Descriptions:** Detailed descriptions of each dish, including traditional origins and modern twists.

4. **Recipe Blog:**

- **Traditional Recipes:** Share recipes of beloved traditional dishes, encouraging visitors to try them at home.
- **Modern Techniques:** Articles or videos on modern cooking techniques that enhance traditional recipes.
- **Chef's Tips:** Insights and tips from the chef on cooking, ingredient selection, and plating.

5. **Gallery:**

- **Food Photography:** A gallery of high-quality images showcasing dishes, the restaurant ambiance, and special events.
- **Customer Experiences:** Photos submitted by customers, highlighting their meals, creating a community.

6. **Events:**

- **Upcoming Events:** Information on special events, such as tasting menus, cooking classes, or wine pairing dinners.
- **Private Events:** Details on hosting private events, including catering options and customizable menus.

7. **Reservations:**

- **Online Booking:** An easy-to-use reservation system for customers to book tables directly from the website.



Regenerate response

Conclusion

Congratulations! You just completed the hands-on lab generative AI applications for IT professionals. In this lab, you explored a few use cases for IT professionals in different roles, including software developers, IT security professionals, and web developers.